Critical Minerals Causal Engine

Appendix — Knowledge Graph Renders

Each render: enriched KG snapshot at the indicated year, with shock origin (dark red node), focal commodity, and 1-hop subgraph. Edges annotated with year-specific PRODUCES / PROCESSES shares. Effective control box (bottom-right) shows the binding stage and percentage.

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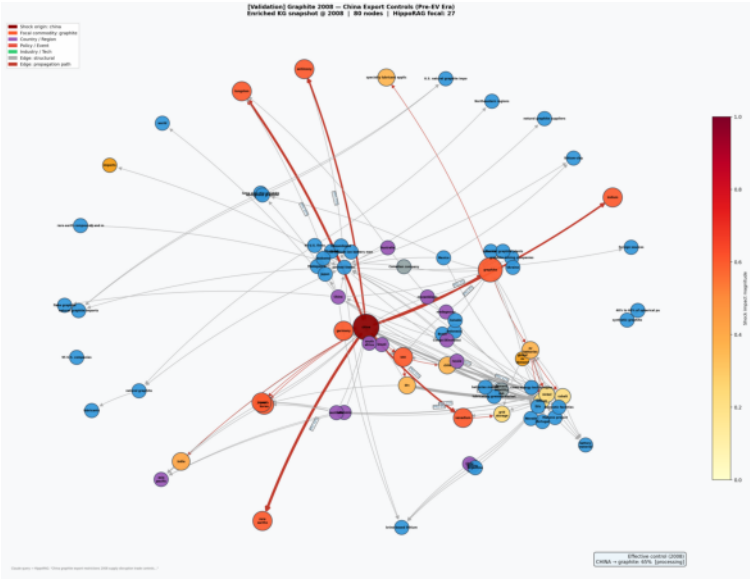
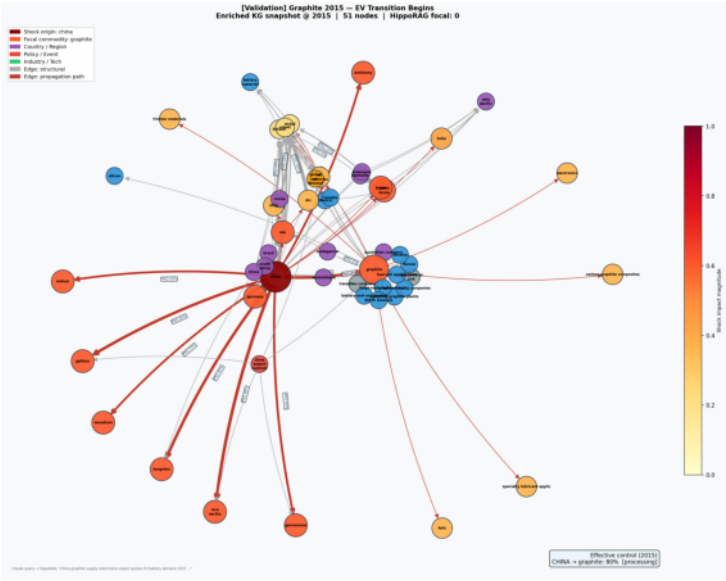
Graphite · 2008



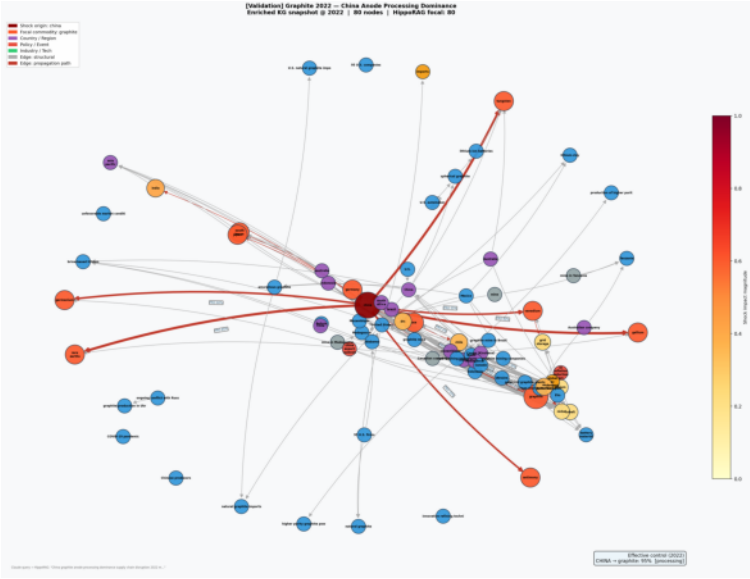
Historical episode — China export quota — pre-EV cycle

Graphite · 2015



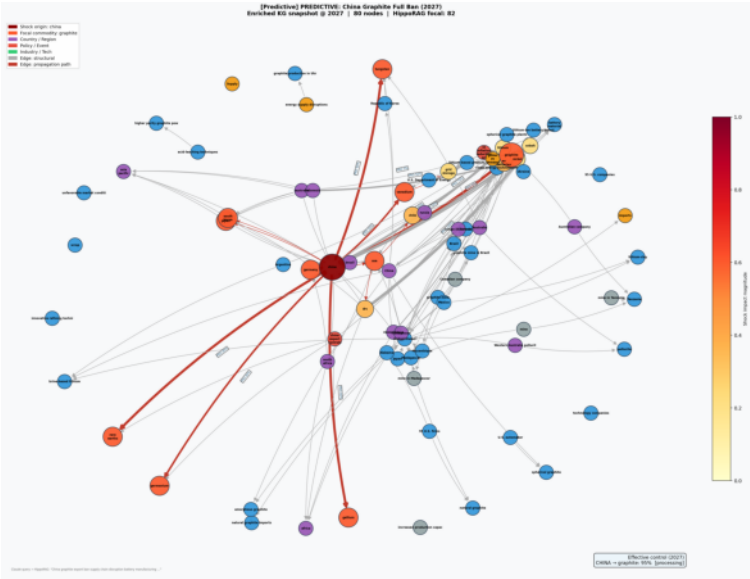
Temporal snapshot — EV transition begins

Graphite · 2022



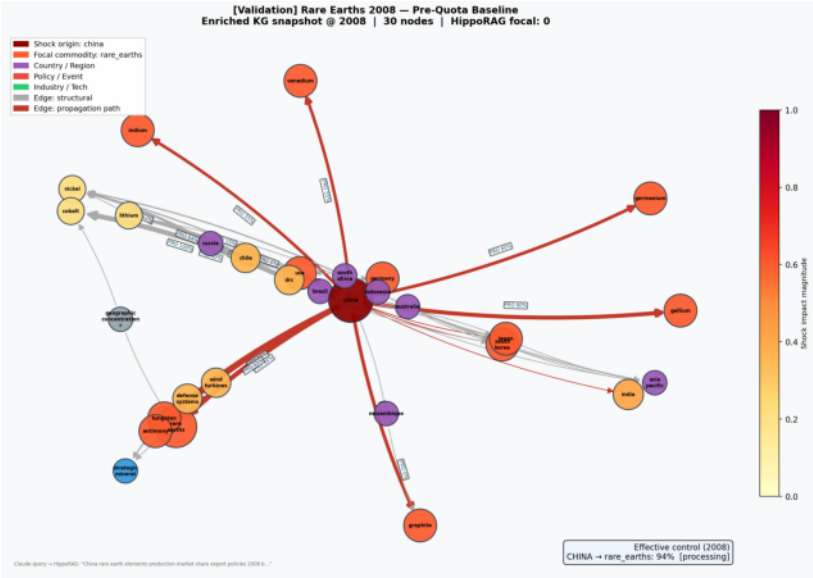
Historical episode — China anode processing 95% lock-in

Graphite · 2027



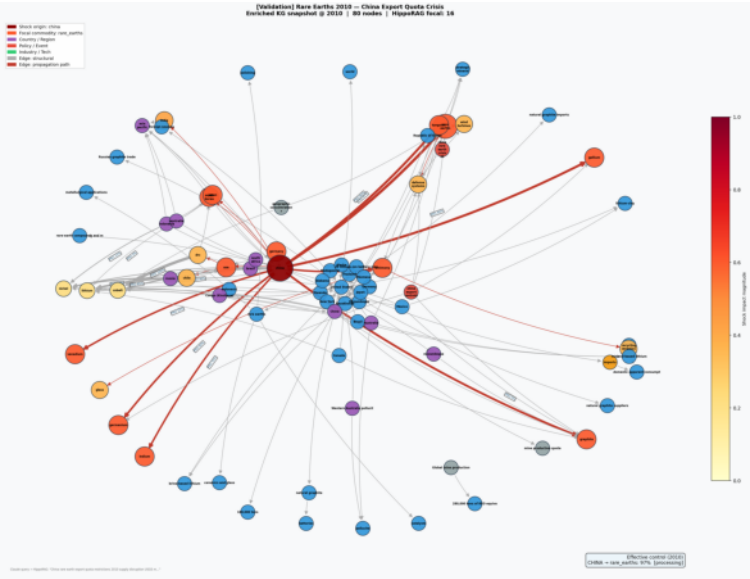
Predictive scenario — PREDICTIVE: full ban scenario

Rare Earths · 2008



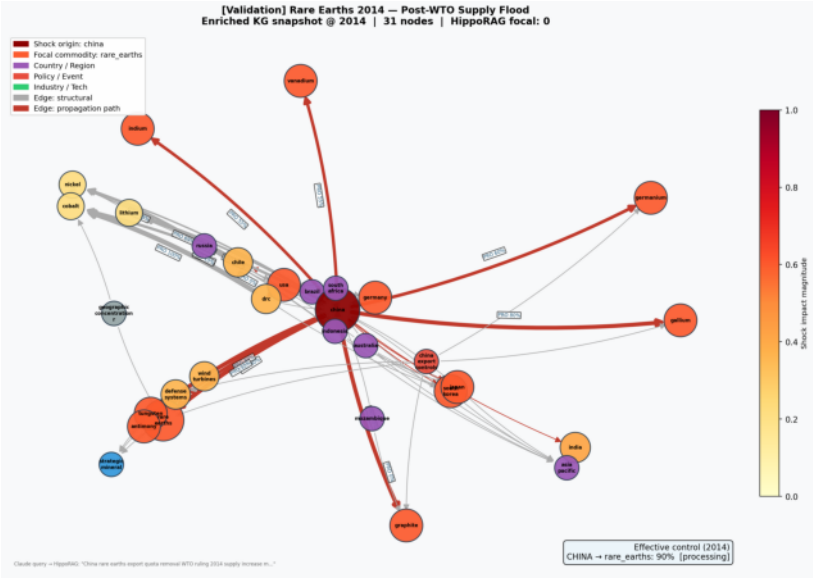
Temporal snapshot — Pre-quota baseline

Rare Earths · 2010



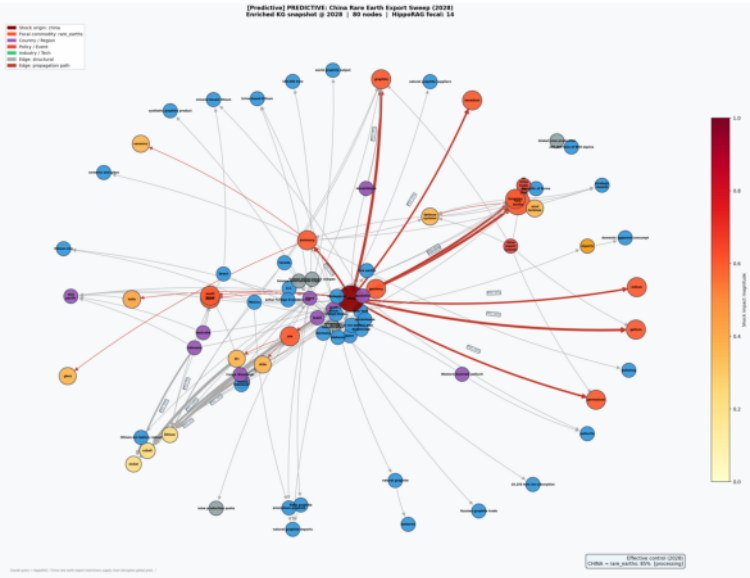
Historical episode — China export quota crisis

Rare Earths · 2014



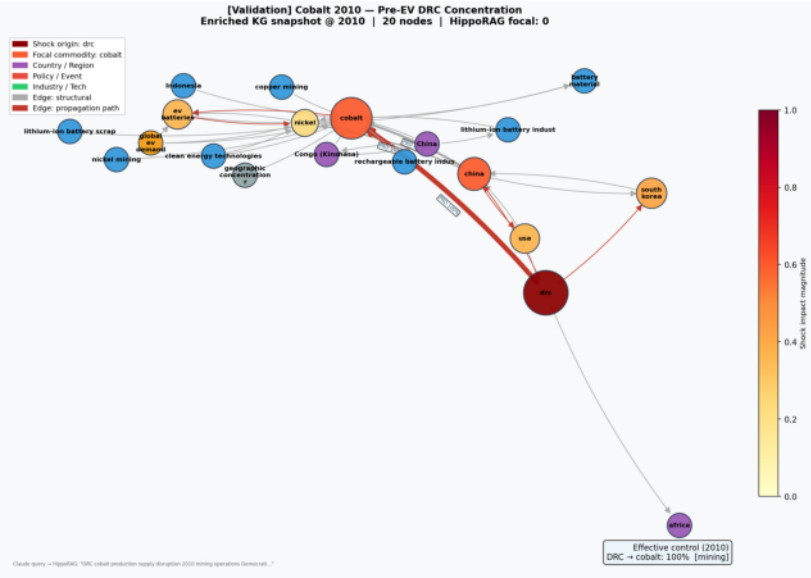
Temporal snapshot — Post-WTO supply flood

Rare Earths · 2028



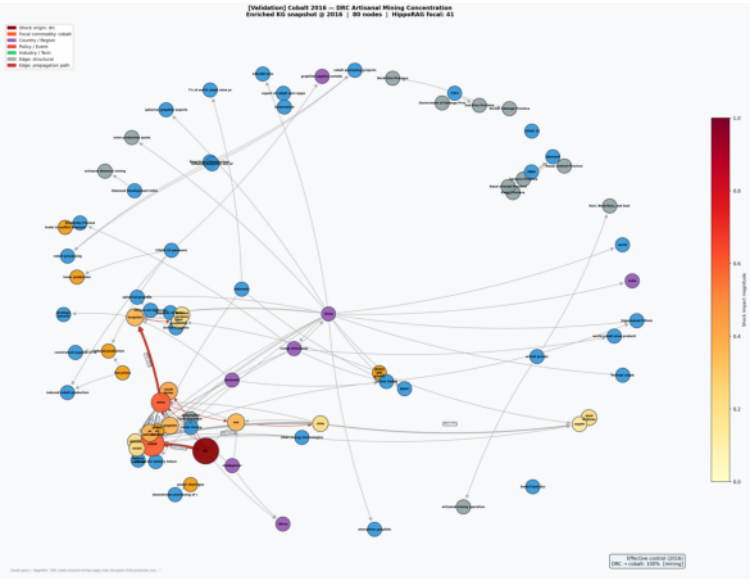
Predictive scenario — PREDICTIVE: export sweep

Cobalt · 2010



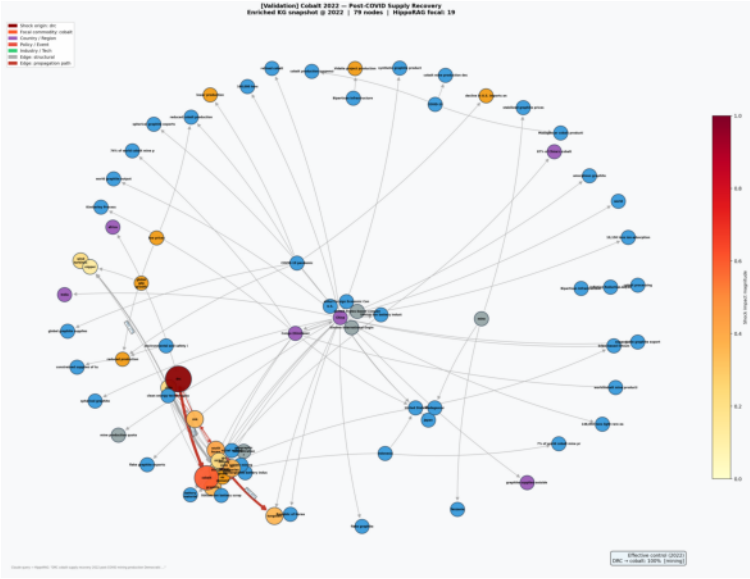
Temporal snapshot — Pre-EV DRC concentration

Cobalt · 2016



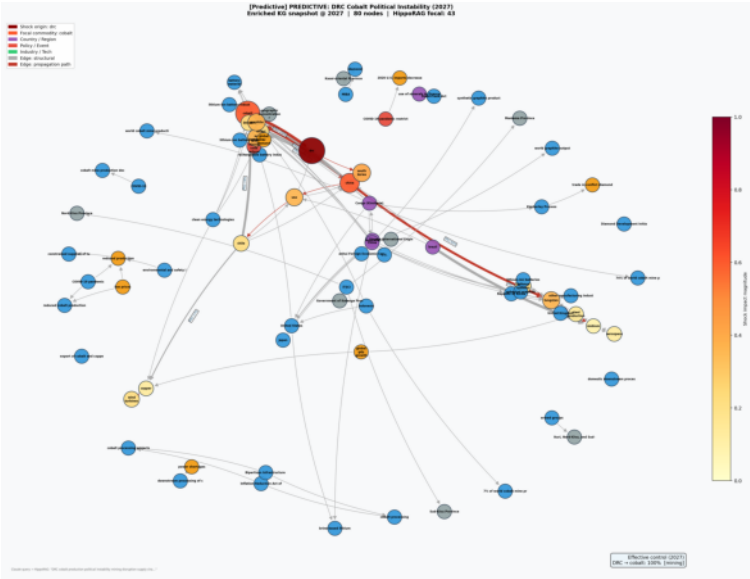
Historical episode — DRC artisanal concentration

Cobalt · 2022



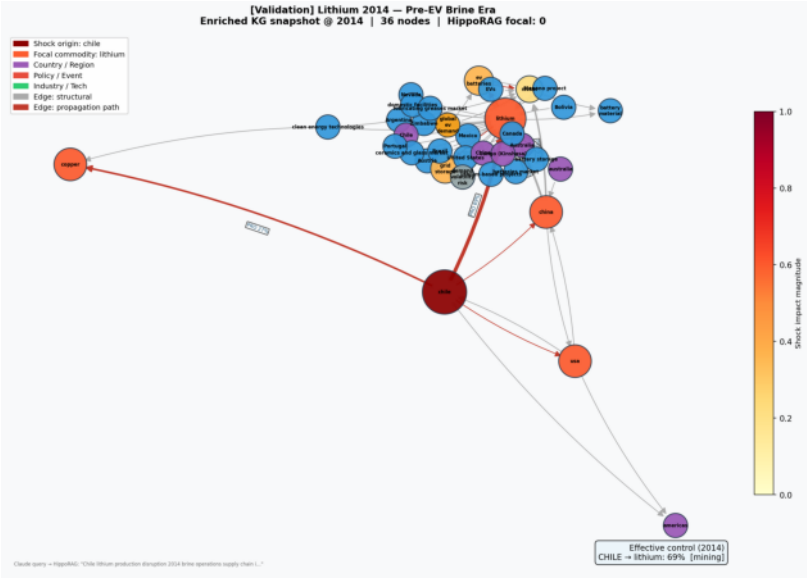
Historical episode — Post-COVID recovery

Cobalt · 2027



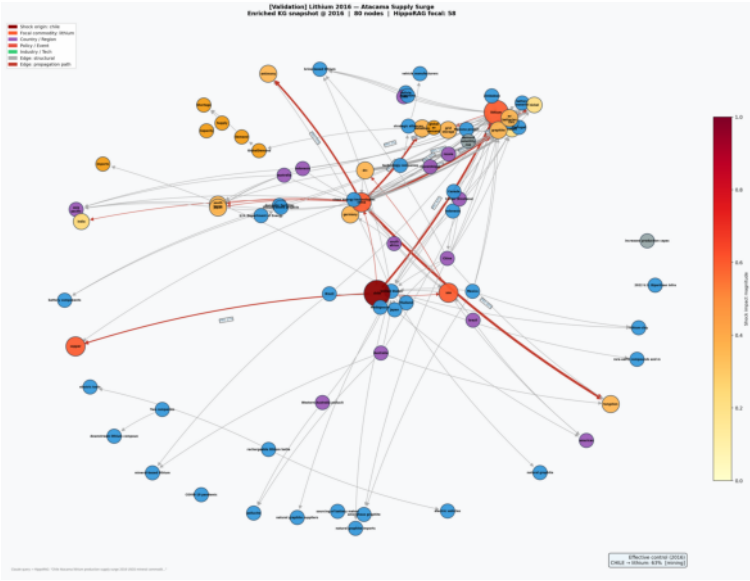
Predictive scenario — PREDICTIVE: DRC instability

Lithium · 2014



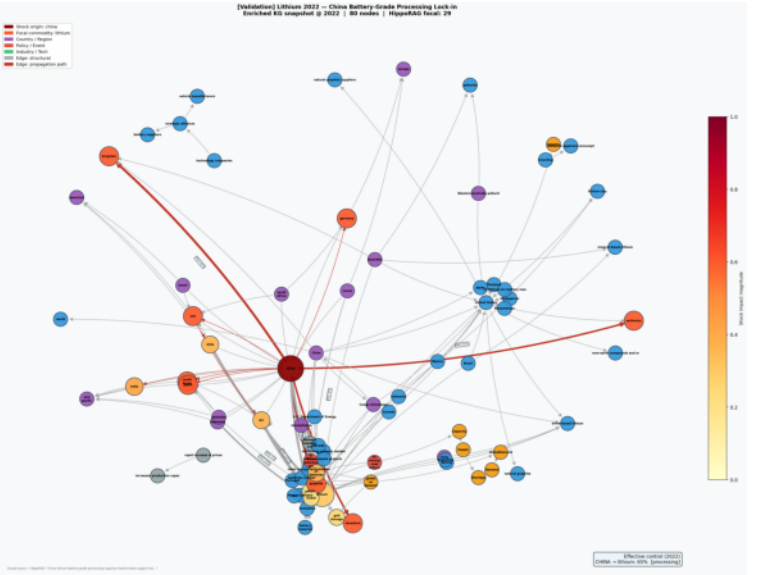
Temporal snapshot — Pre-EV brine era

Lithium · 2016



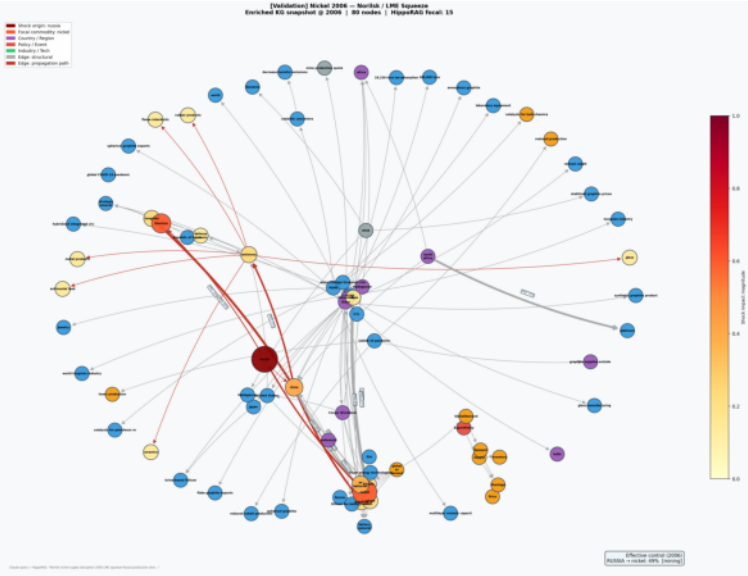
Historical episode — Atacama supply surge

Lithium · 2022



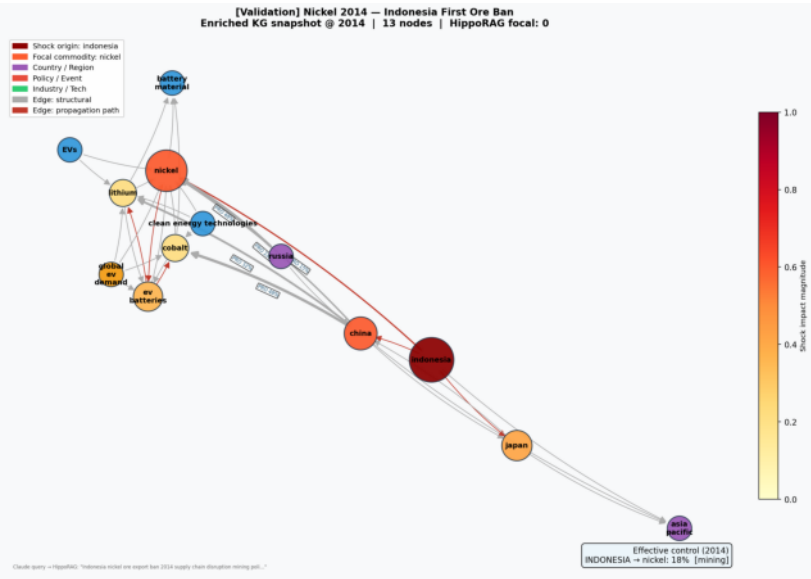
Historical episode — China processing lock-in

Nickel · 2006



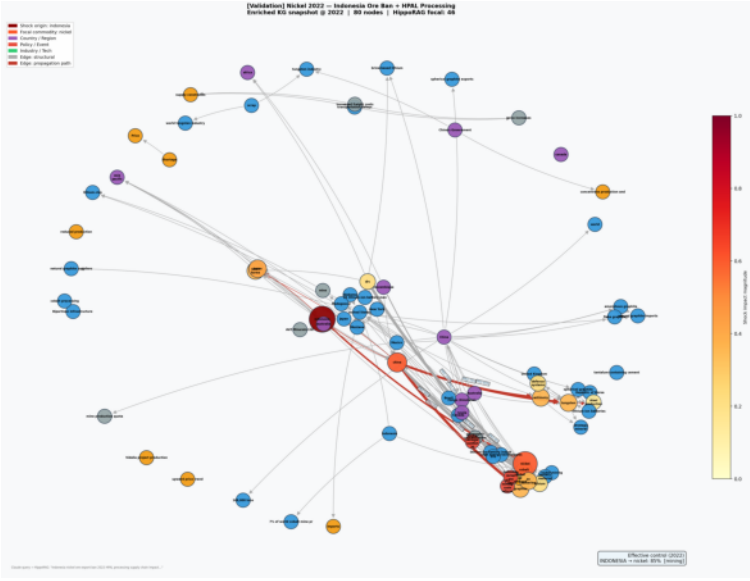
Historical episode — Norilsk / LME squeeze

Nickel · 2014



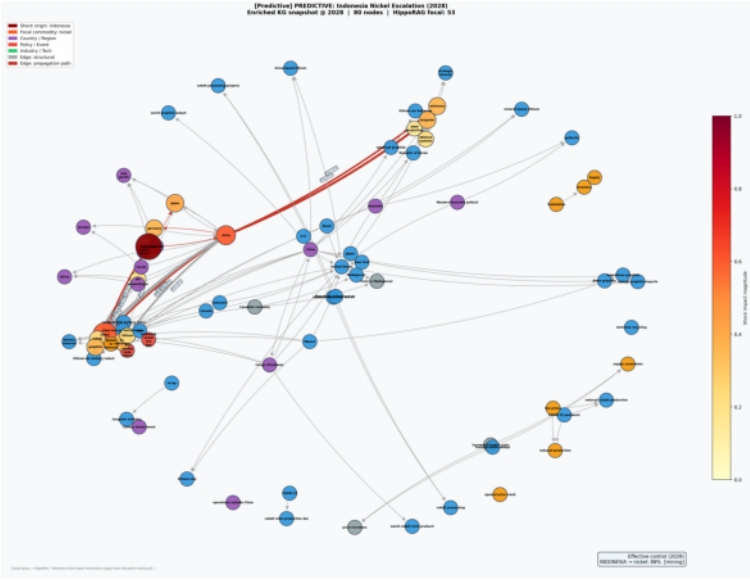
Temporal snapshot — Indonesia first ore ban

Nickel · 2022



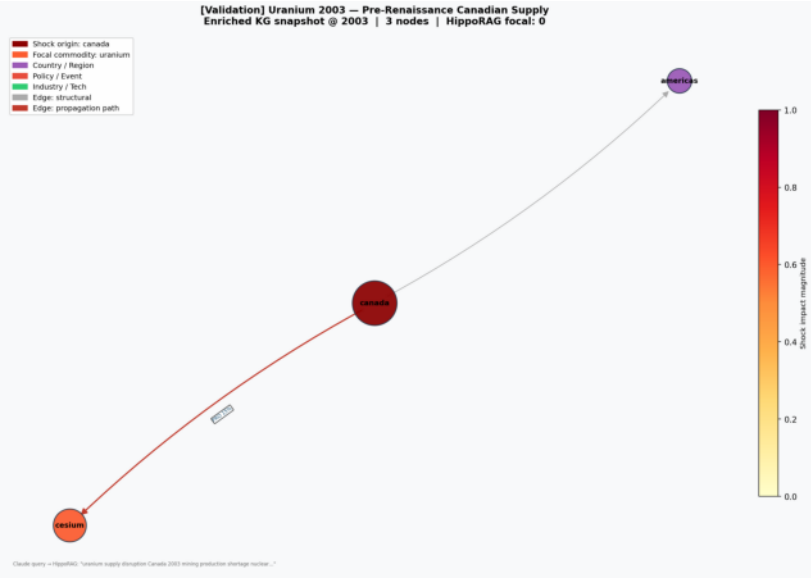
Historical episode — Indonesia ban + HPAL

Nickel · 2028



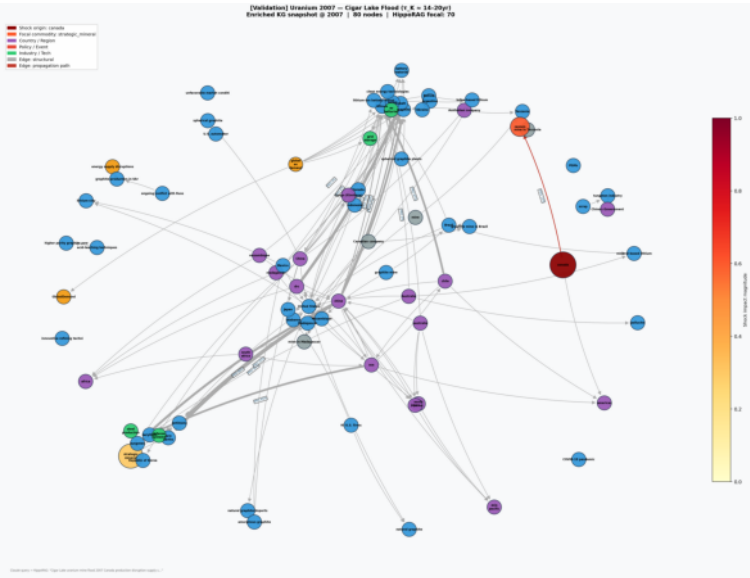
Predictive scenario — PREDICTIVE: escalation

Uranium · 2003



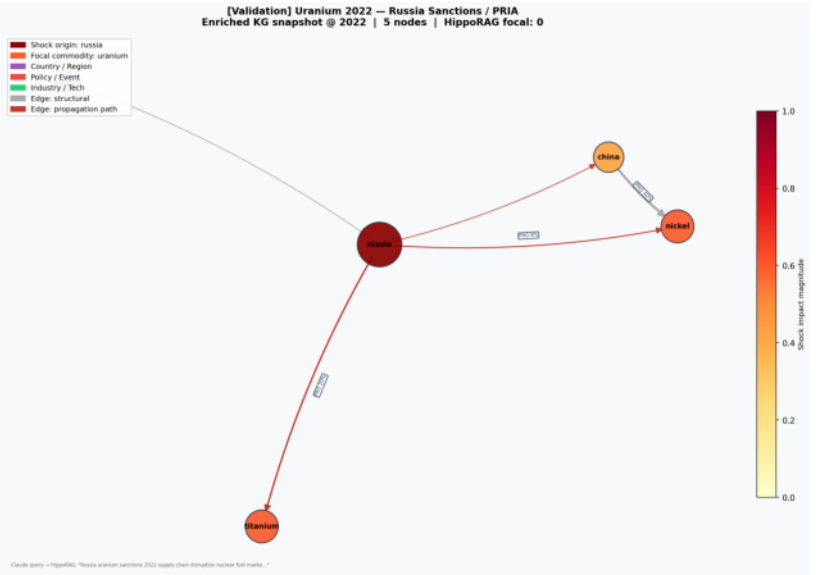
Temporal snapshot — Pre-renaissance Canadian supply

Uranium · 2007



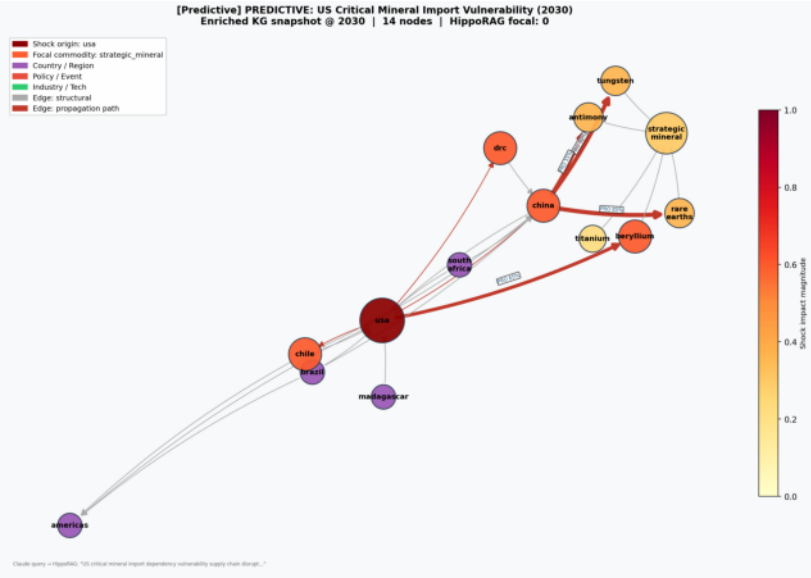
Historical episode — Cigar Lake flood

Uranium · 2022



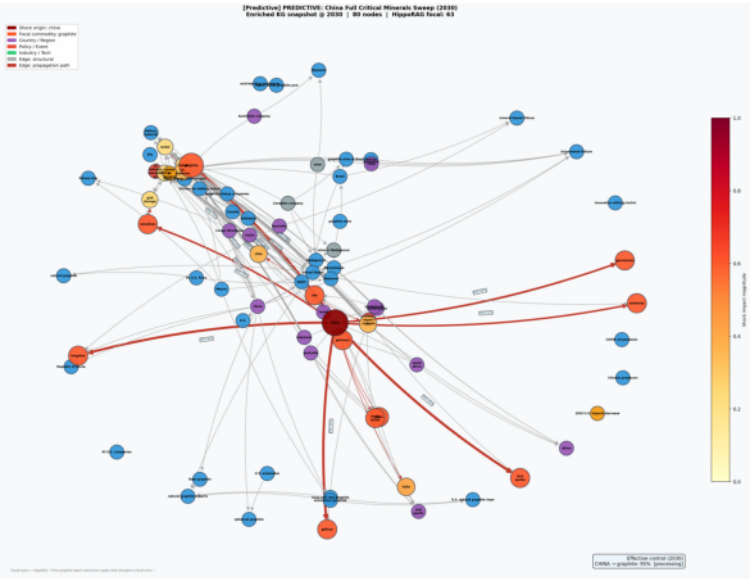
Temporal snapshot — Russia sanctions / PRIA

Cross-cutting predictive · 2030



Predictive scenario — PREDICTIVE: US import vulnerability

Cross-cutting predictive · 2030



Predictive scenario — PREDICTIVE: China full-sweep